

STEP 1 — Install and Import Libraries

```
!pip install transformers==4.37.2
```

```
Collecting transformers==4.37.2
  Downloading transformers-4.37.2-py3-none-any.whl.metadata (129 kB)
    129.4/129.4 kB 10.9 MB/s eta 0:00:00
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.37.2)
Collecting huggingface-hub<1.0,>=0.19.3 (from transformers==4.37.2)
  Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.19.3)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.19.3)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.19.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.19.3)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.19.3)
Collecting tokenizers<0.19,>=0.14 (from transformers==4.37.2)
  Downloading tokenizers-0.15.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.1 kB)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from tokenizers<0.19,>=0.14)
  Downloading transformers-4.37.2-py3-none-any.whl (8.4 MB)
    8.4/8.4 MB 112.1 MB/s eta 0:00:00
  Downloading huggingface_hub-0.36.2-py3-none-any.whl (566 kB)
    566.4/566.4 kB 49.1 MB/s eta 0:00:00
  Downloading tokenizers-0.15.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.6 MB)
    3.6/3.6 MB 88.7 MB/s eta 0:00:00
Installing collected packages: huggingface-hub, tokenizers, transformers
  Attempting uninstall: huggingface-hub
    Found existing installation: huggingface_hub 1.6.0
    Uninstalling huggingface_hub-1.6.0:
      Successfully uninstalled huggingface_hub-1.6.0
  Attempting uninstall: tokenizers
    Found existing installation: tokenizers 0.22.2
    Uninstalling tokenizers-0.22.2:
      Successfully uninstalled tokenizers-0.22.2
  Attempting uninstall: transformers
    Found existing installation: transformers 5.0.0
    Uninstalling transformers-5.0.0:
      Successfully uninstalled transformers-5.0.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are currently installed,
  sentence-transformers 5.2.3 requires transformers<6.0.0,>=4.41.0, but you have transformers 4.37.2
Successfully installed huggingface-hub-0.36.2 tokenizers-0.15.2 transformers-4.37.2
```

STEP 2 — Load Summarization Model

```
from transformers import pipeline

summarizer = pipeline("summarization")

print("Model loaded successfully")
```

```
No model was supplied, defaulted to sshleifer/distilbart-cnn-12-6 and revision a4f8f3e (
Using a pipeline without specifying a model name and revision in production is not recom
/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: FutureWarn
warnings.warn(
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public model
warnings.warn(
config.json:      1.80k/? [00:00<00:00, 86.8kB/s]

pytorch_model.bin: 100%                               1.22G/1.22G [00:08<00:00, 126MB/s]

tokenizer_config.json: 100%                             26.0/26.0 [00:00<00:00, 1.46kB/s]

vocab.json:      899k/? [00:00<00:00, 26.5MB/s]

merges.txt:      456k/? [00:00<00:00, 22.9MB/s]
Model loaded successfully
```

STEP 3 — Prepare Input Texts (Dataset)

```
text = """
Artificial Intelligence is rapidly transforming industries.
Machine learning algorithms help computers learn from data and make predictions.
Many companies are investing in AI to improve efficiency and automate tasks.
"""
```

```
summary = summarizer(text, max_length=40, min_length=10)
```

```
print(summary[0]['summary_text'])
```

```
Machine learning algorithms help computers learn from data and make predictions . Many
```

Step 4 — Generate Summaries for Dataset

```
texts = [
```

```
    """Artificial Intelligence is transforming industries across the world.
    Companies use machine learning algorithms to analyze large datasets and
    make better decisions. AI systems can recognize speech, translate languages,
    and generate human-like text. However, the development of AI raises concerns
    about privacy and job displacement. Experts believe responsible AI
    development is necessary for society.""" ,
```

```
    """Climate change is one of the most serious environmental challenges today.
    Rising temperatures cause melting glaciers, rising sea levels, and extreme
    weather events. Scientists suggest reducing greenhouse gas emissions
    and switching to renewable energy sources like solar and wind power.
    Public awareness and sustainable practices are important for protecting
    the planet.""" ,
```

```
    """Natural Language Processing is a branch of Artificial Intelligence that
    allows computers to understand human language. Applications include chatbots,
    sentiment analysis, language translation, and text summarization. Modern NLP
    systems rely heavily on deep learning and transformer models.""" ,
```

```
"""In modern software development, version control systems such as Git play
an important role. They allow developers to track changes in code,
collaborate with teams, and manage different versions of software.
Version control also helps identify bugs and maintain project history."""
```

```
"""Once there was a small village surrounded by mountains and forests.
The villagers lived simple lives and depended on farming. One day a traveler
visited and shared stories about modern technologies and education.
Inspired by these stories, the younger generation began exploring new
opportunities to improve their future."""
]
```

```
for i, text in enumerate(texts):
```

```
    summary = summarizer(text, max_length=60, min_length=20, do_sample=False)
```

```
    print("\nTEXT", i+1)
```

```
    print("Original Text:", text)
```

```
    print("\nSummary:")
```

```
    print(summary[0]['summary_text'])
```

```
    print("\n-----")
```

```
and generate human-like text. However, the development of AI raises concerns
about privacy and job displacement. Experts believe responsible AI
development is necessary for society.
```

```
Summary:
```

```
Companies use machine learning algorithms to analyze large datasets and make better d
```

```
-----
Your max_length is set to 60, but your input_length is only 53. Since this is a summar
```

```
TEXT 2
```

```
Original Text: Climate change is one of the most serious environmental challenges toda
Rising temperatures cause melting glaciers, rising sea levels, and extreme
weather events. Scientists suggest reducing greenhouse gas emissions
and switching to renewable energy sources like solar and wind power.
Public awareness and sustainable practices are important for protecting
the planet.
```

```
Summary:
```

```
Rising temperatures cause melting glaciers, rising sea levels, and extreme weather ev
```

```
-----
Your max_length is set to 60, but your input_length is only 54. Since this is a summar
```

```
TEXT 3
```

```
Original Text: Natural Language Processing is a branch of Artificial Intelligence that
allows computers to understand human language. Applications include chatbots,
sentiment analysis, language translation, and text summarization. Modern NLP
systems rely heavily on deep learning and transformer models.
```

```
Summary:
```

```
Applications include chatbots,sentiment analysis, language translation, and text summ
```

Version control also helps identify bugs and maintain project history.

Summary:

Version control systems such as Git play an important role in software development .

TEXT 5

Original Text: Once there was a small village surrounded by mountains and forests. The villagers lived simple lives and depended on farming. One day a traveler visited and shared stories about modern technologies and education. Inspired by these stories, the younger generation began exploring new opportunities to improve their future.

Summary:

Once there was a small village surrounded by mountains and forests, the villagers liv

Step 5 — Control Summary Length

```
text = texts[0]

short_summary = summarizer(text, max_length=25, min_length=10, do_sample=False)
medium_summary = summarizer(text, max_length=50, min_length=20, do_sample=False)
long_summary = summarizer(text, max_length=80, min_length=30, do_sample=False)

print("SHORT SUMMARY:\n", short_summary[0]['summary_text'])
print("\nMEDIUM SUMMARY:\n", medium_summary[0]['summary_text'])
print("\nLONG SUMMARY:\n", long_summary[0]['summary_text'])
```

Your max_length is set to 80, but your input_length is only 70. Since this is a summarizer, it will truncate the input text to fit the max_length.

SHORT SUMMARY:

Companies use machine learning algorithms to analyze large datasets and make better decisions.

MEDIUM SUMMARY:

Companies use machine learning algorithms to analyze large datasets and make better decisions.

LONG SUMMARY:

Companies use machine learning algorithms to analyze large datasets and make better decisions.

Step 6 — Very Long Text Test

```
long_text = texts[0] + texts[1] + texts[2]

summary = summarizer(long_text, max_length=120, min_length=50, do_sample=False)

print(summary[0]['summary_text'])
```

Artificial Intelligence is transforming industries across the world . Companies use machine learning algorithms to analyze large datasets and make better decisions.

Dataset Text — AI (Technical), Climate Change (News Type), NLP (Academic), Story (Narrative), Software Development (Technical)

```
text1 = """
Artificial Intelligence (AI) has become one of the most transformative technologies of
"""
```

```
text2 = ""  
Climate change is a global issue that affects ecosystems, economies, and human lives. (  
""
```

```
text3 = ""  
Natural Language Processing (NLP) is a field of Artificial Intelligence that focuses on  
""
```

```
text4 = ""  
In a quiet village surrounded by hills and rivers, people lived simple and peaceful lives.  
""
```

```
text5 = ""  
Software development is a structured process of designing, creating, testing, and maintaining  
""
```

```
texts = [text1, text2, text3, text4, text5]
```

STEP 7 — Compare Text Types

The summarization model performs well for structured texts like news and technical content because it can easily identify key information. For narrative texts, the model captures the main storyline but may remove descriptive details. Technical texts are summarized accurately but sometimes complex terms are simplified. Conversational texts are harder to summarize due to informal language. The model generally maintains coherence but may lose some important details. It performs best on formal and structured content. Overall, the summaries are meaningful but not perfect.